Testing models for the radio emission of x-ray binary systems

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Abstract

Variable radio emission from astrophysical sources is often associated with ejected plasmas that emit radiation at radio frequencies. A standard model used to describe this phenomenon is the 'van der Laan' model [1]. In this project I use a new quantity called the spectral peak index to test this model quantitively against data taken from various x-ray binary systems. I also examine the Time delay between the flux density maxima of different frequencies. The model was found to be a poor predictor of these systems.

I. Introduction

In 1966, Professor H. van der Laan formulated the first model [1] to describe the time-variable radio emissions of synchrotron radiation that had been recorded by contemporary experiments. It had already been proposed that these sources would progress from being optically thick to optically thin at successively smaller wavelengths [2] [3] (optical depth is a measure of the likelihood that a photon will be absorbed when passing through a medium, with "optically thick" representing an opaque object), but van der Laan set out a quantitive model to describe these events that could make testable predictions. This report will describe the origins of the model, and test it against more recently recorded data from a variety of relevant radio sources.

The van der Laan model is designed to predict the behavior of variable extra-galactic radio sources, but is commonly assumed to also apply to x-ray binary systems. These systems consist of one object, either a black hole or a neutron star, absorbing matter from another, usually a star. This accretion releases a large amount of energy, in the form of gravitational potential energy [4]. X-ray binary systems are characterized by relatively brief outbursts with long periods of quiescence in between. These outbursts approach the Eddington limit – the maximum luminosity an object can have while the forces of radiation and gravity remain in balance. Importantly, all black hole x-ray binaries follow the same general time evolution during these outbursts. [5]

The outbursts generally last from a few months to a few years. The outburst sequence is initiated when the mass transfer rate of the donor star is greater than the accretion rate of the black hole. This results in a buildup of mass, and thus a rise in temperature. At a certain point (~4000K) the hydrogen begins to ionize and the viscosity sharply rises, leading to a greatly increased accretion rate. This causes an ejection of matter from the poles of the black hole in the form of a jet, although the underlying mechanism for this is not yet fully understood. One leading theory is that the increased accretion, through the release of gravitational potential energy, generates a much larger magnetic field and thus accelerates the particles. Since the accretion rate

is now larger than the matter transfer rate, eventually the temperature drops, and the cycle begins again. The quiescent phase, as the black hole slowly cools down and heats up again, can last several centuries. I will briefly discuss the relationship between the events' duration and the size of the system later.

The jet itself consists of relativistic electrons accelerated by a magnetic field. This, through the mechanism of synchrotron radiation, is the source of the radio emission we observe. The eponymous x-rays are emitted from the accretion disk itself. Van der Laan made the following prediction of the time evolution of an emitting 'cloud', which is analogous to the jet of the black hole: Initially, the jet is dense, and thus optically thick. Over time, the jet expands, and thus gradually becomes optically thin. Based on the assumption that no further matter or energy is introduced, the energy of the electrons inside the jet decreases as the jet expands. Because of these two opposing influences on the observed flux density, it will at some point be at a maximum. The jet will become optically thin for the higher frequencies earlier, so their observed flux densities will peak earlier than those of lower frequencies; because the internal energy decreases with time, this means that the higher frequencies will also peak higher.

To describe this, in his paper van der Laan considered a spherical cloud of relativistic electrons of radius r. The energy distribution of these electrons is the well-known distribution

$$(1) \quad n(E)dE = K(t)E^{-p}dE \ ,$$

where n(E) is the number of electrons of energy E per cm^3, $K(t)$ is some function of time, and p is a constant known as the electron index. Experimentally, the electron index is nearly always within the range $2<p<2.5$. This fact allows the van der Laan model to be tested against many different systems, as is described below in **II**. Imposing that the source's brightness temperature must not exceed the electron temperature leads to the relation [2] [3]

$$(2) \quad v_m = CB^{\frac{1}{5}}Q^{-\frac{4}{5}}S(v_t)^{\frac{2}{5}}\left(\frac{v_t}{v_m}\right)^{\frac{p-1}{5}} \ ,$$

where v_m is the frequency of maximum flux at a given time, B is the magnetic field that is accelerating the electrons and thus causing them to radiate, Q is the angular diameter of the cloud, $S(v)$ is the flux density of v, and v_t is some frequency for which the cloud is optically thin. Initially, the cloud is assumed to be optically thick for all frequencies. Since the specific emission mechanism being studied is specifically synchrotron radiation, it follows that the frequency dependence of the spectral curve is

$$(3) \quad S(v) \propto v^{\frac{5}{2}}(1 - e^{-\tau(v)}) \ ,$$

where $\tau(v)$ is the optical depth. The time evolution of this power distribution of electrons producing synchrotron emission is the basis of the model. From here, van der Laan made the following assumptions: the magnetic field follows an inverse square law;

particles can neither enter nor escape the cloud; and the gas cools adiabatically. This leads to the equation

$$(4) \quad S(v,\rho) = S_o \left(\frac{v}{v_o}\right)^{\frac{5}{2}} \rho^3 \left(\frac{1-e^{-t_m\left(\frac{v}{v_o}\right)^{-\frac{p+4}{2}}} \rho^{-2p-3}}{1-e^{-t_m}} \right),$$

where $\rho = {}^{r}/_{r_o}$ is the ratio of the current to the initial radius, $S(v,\rho)$ is the observed flux density of a certain frequency at a given time, v_o and S_o are constants representing the theoretical maxima for v and S respectively, and τ_m is the optical depth when v_o is at a maximum. τ_m is the solution to

$$(5) \quad e^x - \frac{p+4}{5}x = 1. \ .$$

For a p of 2.5, t_m has a value of ~0.50. (5) is easily modelled by a computer program:

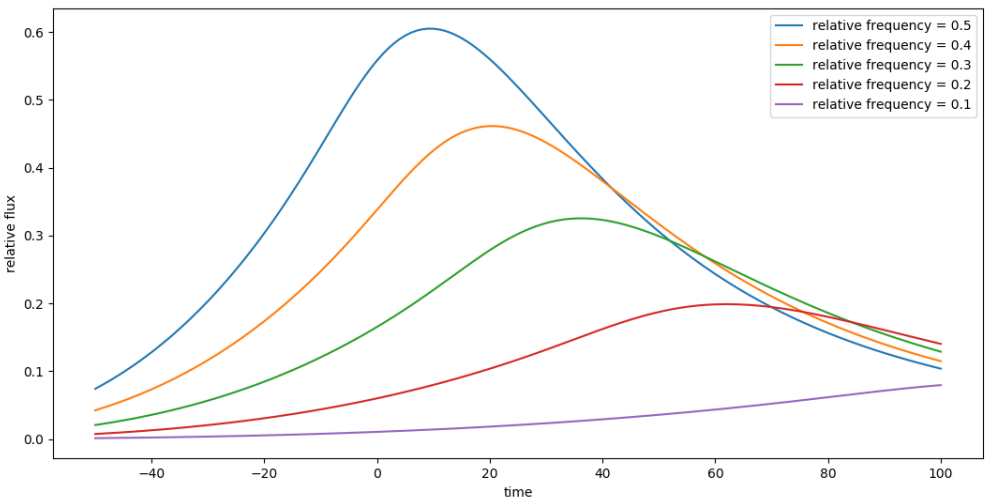

Figure 1 A computer generated diagram of the expected time evolution of a van der Laan event. In this diagram, the electron index p = 2.5, the maximum flux density S_o = 1, and the optical depth for which v_o is maximal t_m = 0.50. Relative frequency must always be < 1, since v_o is the maximum frequency released in the event. Because of the way the van der Laan model is constructed, t = 0 is the time where the highest frequency of the event (relative frequency = 1) is at a maximum.

As is made clear from Figure 1, lower frequencies peak lower and later, as explained above. In addition, their tails are larger than the higher frequencies because the jet is optically thin. This effect is described by the well-known relation for the spectral index in an optically thin cloud:

$$(6) \quad \alpha = \frac{1-p}{2}$$

The spectral index α, a standard metric, should not be confused with the *spectral peak index* (*SPI*) described below. The spectral index is a measure of the ratio of the fluxes of two different frequencies at a specific time, while the *SPI* is a pseudo-spectral

index from measuring the peaks of two different frequencies, which may happen at different times. For $p = 2$, the spectral index $\alpha = -0.5$, meaning that the lower frequencies will have higher flux than the higher ones.

Here, I test this model against the evolution of radio emission from some black hole x-ray binaries. Qualitatively the model seems to agree with the general structure of what is observed (higher frequencies peak higher and earlier). However, it has often been noted anecdotally that the van der Laan model does not always accurately predict the relative peak fluxes of different frequencies. Here it will be shown that the model does not make reliable quantitive predictions for x-ray binaries, and thus is not a robust model for these systems.

II. Method

We can test the van der Laan model by looking at the relationship between the peak fluxes of two different frequencies from the same event. Van der Laan gives this equation for the frequency that is peaking at a certain time:

$$(7) \quad v_m(\rho) = v_o \rho^{-\frac{4p+6}{p+4}} \ .$$

Van der Laan appears to define ν_m as the frequency with the maximum value at a given time, but upon closer examination of his mathematics one realises that he does in fact mean the frequency that is reaching its maximum at a given time. This can be rewritten as:

$$(8) \quad \rho = \left(\frac{v_m}{v_{mo}}\right)^{-\frac{p+4}{4p+6}}$$

Within the van der Laan model, if a frequency is peaking, then it is perforce the frequency with the highest flux at that time, v_m. Therefore, we can use (8) to simplify (4):

$$(9) \quad S_m(v) = S_o \left(\frac{v}{v_{mo}}\right)^{\frac{5}{2}} \left(\frac{v_m}{v_{mo}}\right)^{-\frac{3p+12}{4p+6}} \left(\frac{1-e^{-t_m\left(\frac{v}{v_{mo}}\right)^{-\frac{p+4}{2}}\left(\frac{v}{v_{mo}}\right)^{\frac{p+4}{2}}}}{1-e^{-t_m}}\right) = S_0 \left(\frac{v}{v_{mo}}\right)^{\frac{7p+3}{4p+6}} \ .$$

Since p reliably takes a value of $2<p<2.5$, we can easily compare events from different systems by taking the *spectral peak index*:

$$(10) \ SPI = \frac{\ln\left(\frac{S_m(v_1)}{S_m(v_2)}\right)}{\ln\left(\frac{v_1}{v_2}\right)} = \frac{7p+3}{4p+6} \ .$$

This has a range of $1.21<SPI<1.28$ for the typical p range. This is a new tool for testing the van der Laan model in a simple way. Throughout this article a subscript of

"1" refers to the higher frequency. The *SPI* has been calculated for several different systems from data taken from various papers.

Object	***SPI***	***SPI error***
V404	-1.08	0.01
	-0.81	0.00
	-0.272	0.112
	-0.052	0.026
	0.000	0.348
	0.009	0.109
	0.012	0.092
	0.049	0.034
	0.065	0.085
	0.093	0.044
	0.152	0.057
	0.185	0.046
	0.207	0.051
	0.209	0.045
	0.256	0.026
	0.257	0.037
	0.312	0.062
	0.361	0.031
	0.417	0.038
	0.505	0.084
	0.511	0.043
	0.688	0.120
	0.715	0.238
	0.922	0.052

Object	***SPI***	***SPI error***
XTE J1752−223 [6]	-0.93	0.22
	-0.81	0.28
	-0.55	0.48
	-0.50	0.29
	-0.38	0.42
	0.04	0.29
	0.17	0.15
HR95	-0.76	0.09
	-0.55	0.13
GX 339-4 [7]	-0.63	0.41
GRO J1655-40 [8]	-0.50	0.11
XTE J1550-564 [9]	-0.25	0.24
Cyg X-3 [10]	-0.36	0.12
	0.05	0.08
	0.21	0.13
	0.47	0.14
	0.72	0.12
Cir X-1 [11]	0.38	0.34
	0.49	0.36
CI CAM	0.76	0.18
GRS 1915+505 [12]	1.57	0.57
	1.57	0.57

Table 1 A table of the *SPI* values that was calculated from data of various systems. The data is grouped by system, and sorted within their respective systems by value.

These values generally had to be taken by eye from the graphs present in their respective papers. In the cases of V404, HR95, and CI CAM, the values were taken from data sets provided by R. Fender. Because this process was done by eye, errors were estimated conservatively.

The time delays (ΔT) between the peaks were also measured. ΔT was always measured from the peak of the higher frequency to that of the lower frequency. This can also be tested against the model, although it is somewhat more complicated than for the flux ratio. Alongside (7), van der Laan gives us the expression:

$$(11)\quad \frac{S_m(v_1)}{S_m(v_2)} = \left(\frac{\beta t_o + t_2}{\beta t_o + t_1}\right)^{\frac{\beta(7p+3)}{p+4}} ,$$

where the cloud's radius's time evolution is taken to be $\rho = \left({}^{t}/_{\beta t_o}\right)^{\beta}$ and $t_o = \frac{r_o}{\dot{r}_o}$ is the *apparent age*. This can be rewritten in terms of the *SPI*:

$$(12)\quad \frac{\ln\left(\frac{\beta t_o + t_2}{\beta t_o + t_1}\right)}{\ln\left(\frac{v_1}{v_2}\right)} = \frac{SPI}{\beta}\frac{p+4}{7P+3} = \frac{p+4}{\beta(7p+3)} .$$

The complications arise because β and t_o are parameters of the individual system. It is usually assumed that they relate to the size scale of the jet; accordingly, I plotted the time delay against $S_m(v_1)$, another metric for the size of the jet, to test this (Figure 3).

III. Analysis

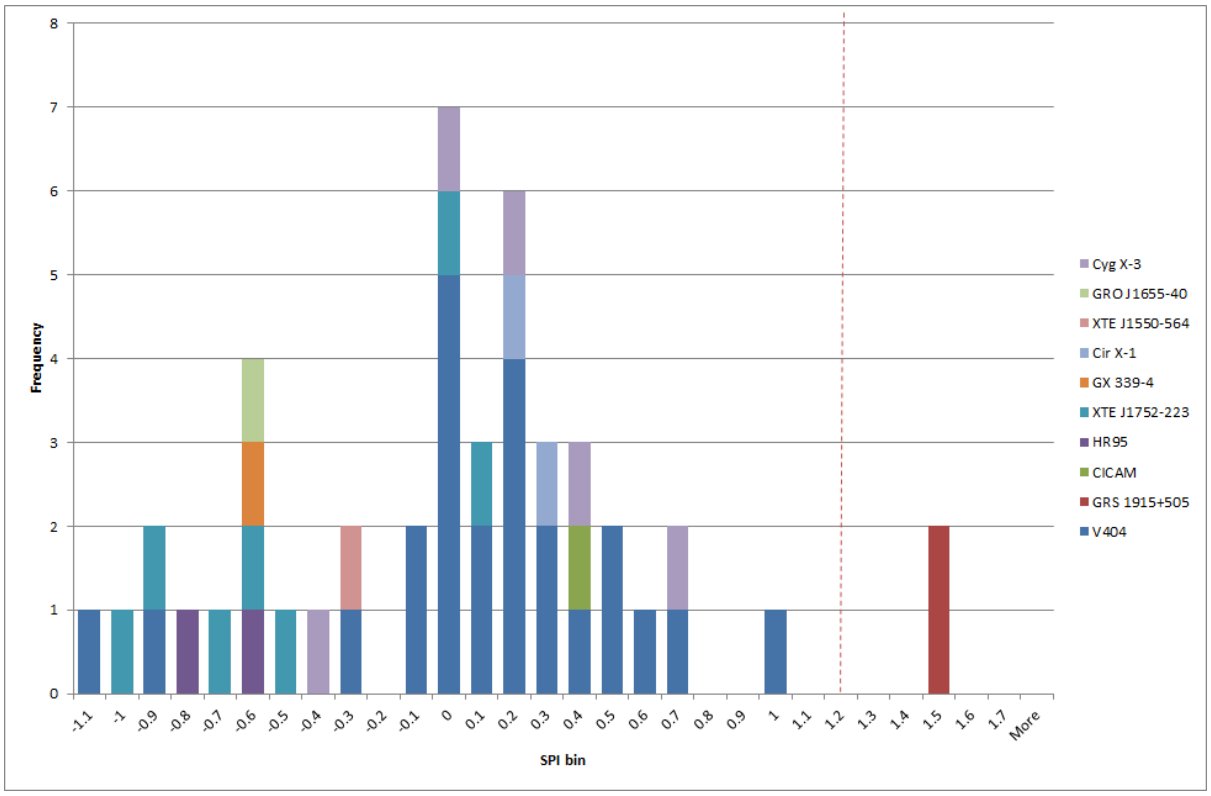

Figure 2 A histogram of the Spectral Peak Indices (10) from the data collected. The dotted red line marks the bin for the SPI predicted by the van der Laan model (for 2<*p*<2.5, which accounts for all observed systems of this kind)

Most events are clustered in the range of -0.1<*SPI*<0.8, below the expected range of 1.21<*SPI*<1.28, marked by the dotted red line. While the data there is inflated by a large amount of data from one system (V404), they are still individual events and are thus relevant to discussion. The events for which *SPI*>0 demonstrate similarities to the van der Laan model, in that the higher frequency does peak earlier, but by a smaller ratio than predicted. So, while the general structure of the model is there, the details

are not. This could be caused by a different rate of change of optical depth than what the model expected.

The negative values are events where the higher frequency peaks lower than the lower frequency. These are events where the jet is optically thin. The spike at *SPI* =-0.6 is particularly notable because -0.6 is the expected value for a *perfectly optically thin* event – an emission where the jet is transparent for the entire duration. An important realization about these events is that they must have prolonged particle acceleration during the rise phase of the event. The maxima in the van der Laan model occur when the increasing transparency is balanced out by the decreasing internal energy. These events are optically thin to start with, so the observed flux should decrease for the whole duration – however, they still demonstrate maxima, so something else must be causing a rise in observed flux, and the only explanation is a prolonged period of particle acceleration. The most likely cause for this is that matter continues to be injected into the jet after the initial outburst, although other causes such as an increasing magnetic field could be the case.

The two events from GRS 1915+505 are within estimated errors of the expected value, and so are close enough that they could be considered to fall within the model. However, the fact that only two events from a total of 39 studied appear to agree with the model suggests that it is not applicable to most events.

Time delay is harder to analyze because there is a normalization factor that varies from system to system(11), and so, if we are to test for a correlation, only one system should be compared at a time. Because of the large number of events from V404 it is an ideal choice:

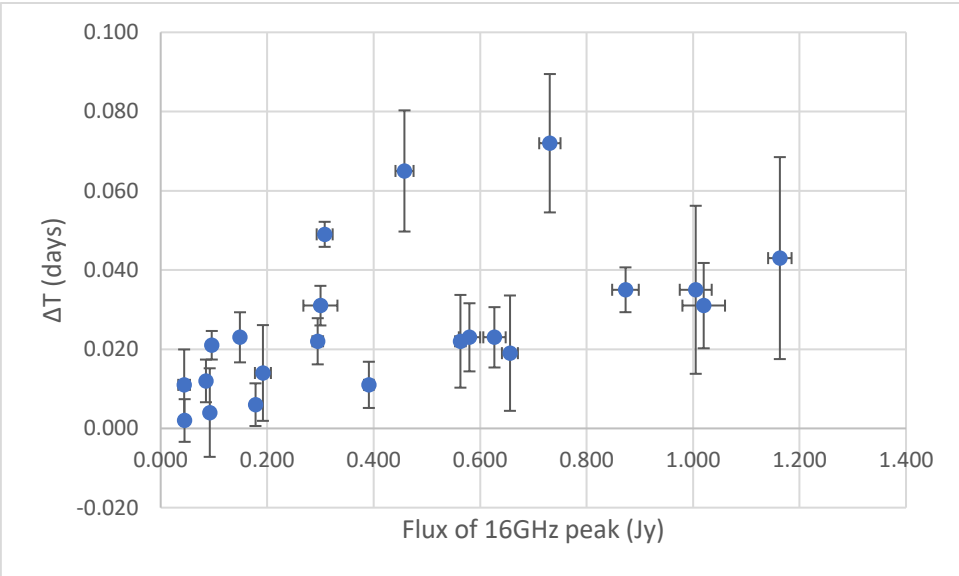

Figure 3 Time delay between peaks of V404 against the magnitude of the first peak. For the sake of readability, the data points corresponding to *SPI* = -0.81 and -1.08 (see Table 1). Both exhibited a substantially negative time ΔT (-0.193 and -0.144 respectively), and were thus considered to be outliers for the sake of this graph.

While superficially there does appear to be a trend, the number of outliers means that we cannot reliably draw any conclusions from this about the relationship between the time delay and the size scale of the jet.

For posterity, here are the time delays from the other events:

Object	***SPI***	***SPI error***	***ΔT (days)***	***ΔT error***
XTE J1752−223	-0.929	0.216	0.000	1.414
	-0.811	0.283	0.000	2.236
	-0.552	0.482	-4.000	1.414
	-0.496	0.292	0.000	1.414
	-0.378	0.418	0.000	1.414
	0.035	0.289	0.000	1.414
	0.166	0.146	0.500	1.414
HR95	-0.757	0.095	1.200	0.141
	-0.549	0.128	-0.800	0.608
GX 339-4	-0.632	0.407	0.004	0.012
GRO J1655-40	-0.503	0.105	-0.360	0.802
XTE J1550-564	-0.251	0.242	-0.520	0.600
Cyg X-3	-0.360	0.125	1.000	11.314
	0.045	0.078	0.000	11.314
	0.209	0.130	0.000	11.314
	0.468	0.136	0.000	11.314
	0.717	0.124	1.000	11.314
Cir X-1	0.377	0.341	0.000	1.414
	0.494	0.363	0.000	1.414
CI CAM	0.755	0.176	26.000	7.071
GRS 1915+505	1.566	0.573	0.004	0.002
	1.566	0.573	0.004	0.002

Table 2 of the time delays of the various events, by system. The systems are sorted by their average *SPI*, as within that are the individual events.

The errors were judged by eye and taken very conservatively. ΔT errors were calculated from uncertainty in the actual times of the peaks (due to eye measurements) using the standard additive formula, so they exist even in cases where the time delay itself is clearly zero. However, they remain useful because they represent the time scale of the events. For example, an error of 1.41 means the events had an estimated 1 day of

error for each peak, based on the scale used. This should be taken into account when considering the veracity of these measurements, since the time scale on the graphs were often not detailed.

There is no apparent relationship between the time delays and the *SPI*s across the systems. This is mostly unsurprising as it supports the statement that β and t_o are parameters of the object itself, and thus are not consistent across different objects. This is further supported by the large range in magnitudes of the time delays.

Events with delay $\Delta T=0$ are expected to be optically thin events. This is because the period of prolonged particle acceleration affects all frequencies concurrently, and thus all maxima are expected to happen at the same time. In events which are not optically thin, the different frequencies are affected at different rates, and so the events should always demonstrate a nonzero time delay. The results of Cyg X-3 and Cir X-1 appear to disagree with this; however, this could be due to the unreliability of measurement. Because the scale of the time delay is unpredictable without knowing the values of β and t_o, the time delay could be sufficiently small in relation to the figure such that it would be measured as null in relation to the graphs taken.

This contradicts the predictions made by the van der Laan model, as do the negative time delays, which represent events where the lower frequency peaks earlier. As negative time delays are exclusive to optically thin events, this is likely a symptom of the model's failure to account for optically thin systems. Finally, there are only three events where the lower peak occurs earlier than the higher peak. This suggests that van der Laan's prediction that the higher peak happens first is reliable in most cases, even for events that are optically thin.

IV. Conclusion

The van der Laan model predicts an *SPI* range of $1.21<SPI<1.28$. As only 2 out of 39 events studied fall within this range, I have demonstrated that the van der Laan model is insufficient to fully describe most black hole x-ray binary systems. Most events have a smaller peak flux ratio than expected, and the model fails to describe why many have the lower frequency's observed flux density peaking higher, which is in direct opposition to the model. This is because the model assumes that the jet is initially optically thick and becomes optically thin over time. Furthermore, the model also fails to explain why some events have no, or even negative, time delay between peaks.

I also investigated how the time delay between the peaks of two different frequencies might relate to the size scale of the jet, but no conclusions could be drawn.

The relation between time delay and event magnitude was also investigated, but no conclusions could be definitively drawn. A model that adequately describes these events is yet to be found. It is likely that a separate model will be required for the optically thin events than for the standard events.

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